Exploring House Prices in Ames, Iowa

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1. Introduction

In this project the Ames housing dataset was used, which contains data about 2930 residential property sales that happened in Ames, Iowa. Each row record contains a set of features describing the different aspects of the house, along with some information about its sale. I answer some questions using this dataset, such as:(1) What's the impact of different features on the houses' price? (2) Which features are strongly connected to the price of the house? (2)

2. Cleaning Summary

The raw dataset had several problems that needed fixing before analysis:

- **Missing values:**
  After examining the missing values in each column, i have found 6 columns which were mostly null, so i dropped them (Alley, Mas Vnr Type, Fireplace Qu, Pool QC, Fence Misc Feature).
- **Data types:** As for the datatypes, I didn't find any columns with the wrong type.
- **Outliers:** After using the boxplot chart for the "SalePrice" column, I have found a lot of outliers, but I didn't remove them because they still are real sales that happened.
  After that, I took the values above the 99th percentile of the data, then capped them.
- **Duplicates:** No duplicates have been found.
  After that, I did the three checks: 1 - No nulls, 2- No negatives, 3- Complete cols (i only dropped the 6 mentioned above)

Then i made a function named "clean_data" I put all the previous steps to clean a database in that function.

3. Feature Engineering Summary

I made some new columns to get new information, Such as:

- Price_per_SF = SalePrice / LotArea. This variable helps us see how much one sqft costs. Lot Area can't be zero because drop all nulls in the dataset.
- TNumOfBathrooms = In this variable, I combined the nums of all bathrooms in each house.
  - Qual x price = This creates an interactive variable
  - SalePrice_log1p = After seeing that the original data was skewed, I transformed the data to remove the skew.

I also one-hot encoded two columns, 1- the Neighborhood column (28 categories) and the Sale Condition columns (6 categories) and ordinally encoded Exter Qual (1-3 scale). Also, I scaled 2 columns: (Lot Area) (Total Bsmt SF) by standardscaling, so its' mean = 0 and std = 1

4. Key Findings

Finding 1 — Overall Quality has the strongest correlation with saleprice. The correlation matrix(fig 1) represents that the column that has the most correlation with the price of the house was the overall quality.

Finding 2 — The years matter. After making a scatter plot that combines the saleprice and overall quality and the year the house was built(fig 2), it has come to me that with time, the house prices in Ames, Iowa has increase, also the overall quality of the houses

Finding 3 — Zones the house is in significantly affect the price! the grouped boxplots, containing the different zones of Ames, Iowa and the Saleprice of the houses(fig 3), has shown that their is a huge difference in house prices depending on the neighborhoods.

Finding 4 — How streets matter. A group box plot that has the saleprice and street material as its axises has been made(fig 4), and it seems that the material of the street the house is on majorly affects its price, showing that the houses on the paved street are significantly more expensive than the houses on gravel.

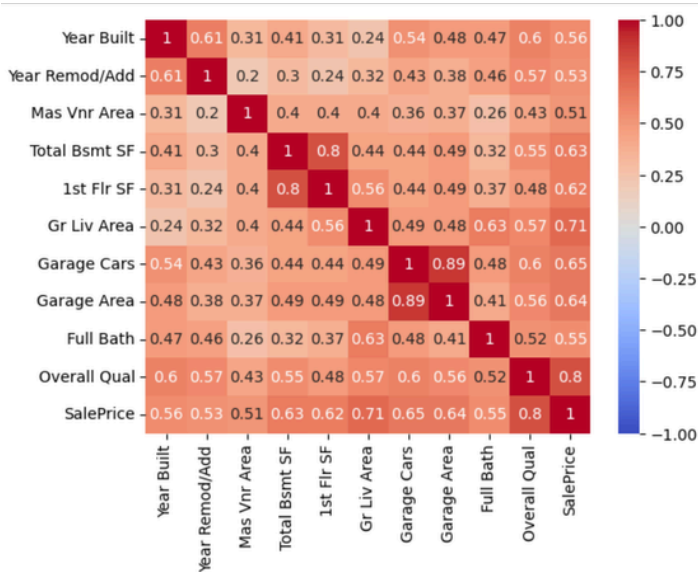

fig 1

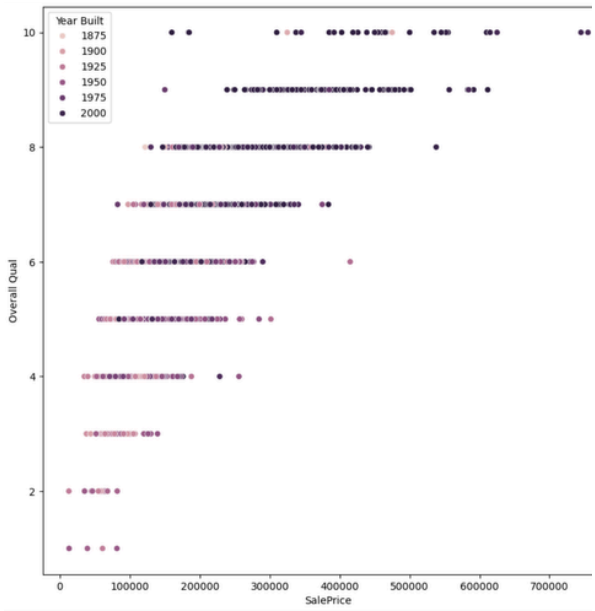

fig 2

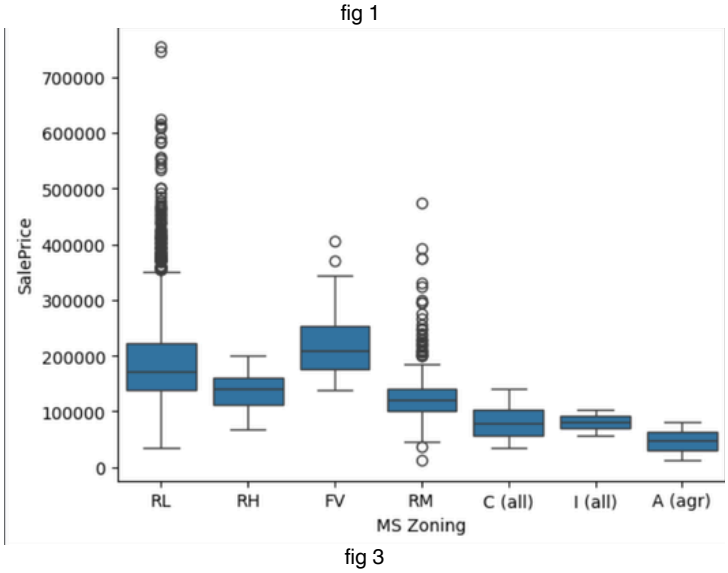

fig 3

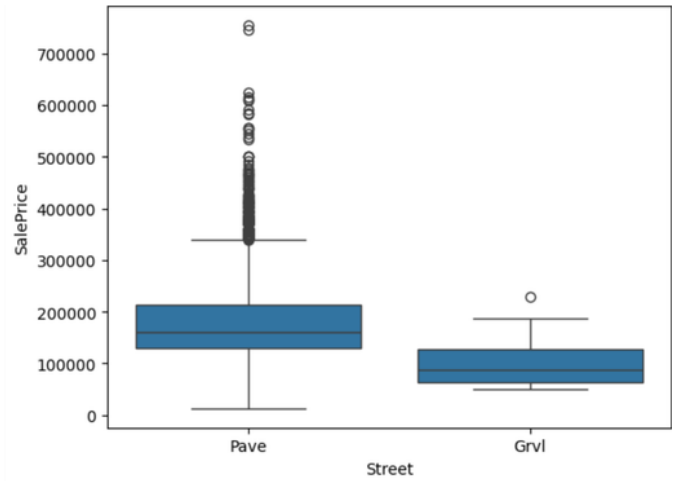

fig 4

5. What I Would Do Next

If I had more time, I would: (1) train a simple linear regression model using some features such as the Lot Area or the Total num of bathroom to predect the house prices; (2)see if the affect of num of bathrooms or num of total rooms affect the house price diffrently base on the neighborhood; (3) see how the diffrent materials the different parts of the house are made of affect the price; (4) see if the num of cars the garage holds is associated with the area of the house